

Prayag Raj Singh

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Education

- University of South Florida, Tampa, FL** Master of Science- Artificial Intelligence and Business Analytics **Aug 2024 – May 2026**
• **Coursework:** Machine Learning, Stats, Database Management, Big data for business, Advance Data science, Data warehousing
- University of Mumbai, Mumbai, India** Bachelor of Engineering- Electronics Engineering **Jun 2018 - Jun 2022**

Skills

- **Programming & Databases:** Python, Java, SQL, R, PostgreSQL, Oracle, Snowflake, CassandraD, SQLAlchemy, FASTAPI, Microservices
- **Data Engineering:** Kafka, Airflow, Mage, ETL/ELT, Docker, CI/CD, PySpark, Hadoop, Map-reduce, Hive, Liquibase, Iceberg
- **Cloud & Big Data:** AWS (S3, Glue, Redshift), GCP (BigQuery), Databricks, Cloudera, MLlib, terraform.
- **Analytics & Visualization:** Tableau, PowerBI, Looker, NumPy, Pandas, Machine Learning, GenAI, XGBoost, streamlit

Related Experience

Depository Trust & Clearing Corporation (DTCC), Tampa | *IT Intern (Data engineering)* **May 2025 - Dec 2025**

- Developed the backend of a Python-based financial data synthesizer, enabling secure generation of synthetic financial datasets for usage
- Integrated Oracle Database to support structured querying and scalable storage of data used by multiple engineering and analytics teams.
- Pipelined synthetic financial data to support data science use cases using snowpipes, enabling safe and scalable ML experimentation.
- Accelerated team delivery by contributing to a test data-as-a-service platform, reducing manual dataset prep time by 40%.
- Engineered a CI/CD pipeline for automated OracleDB data migrations across on-prem and cloud environments using Liquibase and Jenkins, improving reliability and increasing 70% deployment speed.
- Migrated **2,000+** transactional database components and **5TB** of data, reducing manual efforts and accelerating delivery by over **6 months**

University of South Florida | *Teaching Assistant (Data Science)* **Sep 2024 - May 2025**

- Guided 300+ students in applying data science principles using Tableau and python, enhancing their ability to derive insights from data.
- Engineered and maintained a master SQL database to ensure accurate, organized data management for course operations.
- Developed Python and machine learning scripts to automate certificate issuance and detect duplicates, streamlining workflows.
- Delivered technical presentations on Tableau and Alteryx advanced features fostering data-driven decision-making and visualization skills.

Navneet Education Limited & Global Edutech | *Product Data Analyst* **Jan 2023 - Jul 2024**

- Managed a portfolio of six LMS and ERP products, accelerating feature deployment by 30% through data-driven development
- Designed data pipelines using SSIS and SQL server, reducing data processing time and enabling insights for data-driven decisions
- Integrated Razorpay payment gateway into ERP, automating 10,000+ monthly transactions and improving workflow efficiency
- Automated ETL workflows, increasing data accessibility by 50% and reducing reporting latency
- Collaborated cross-functionally (engineering and data science) to engineer data pipelines, cutting report generation time by 40%

Metron Security | *Software Data Analyst* **Oct 2022 - Jan 2023**

- Built data pipelines with Python (NumPy, Pandas) and optimized SQL, improving database performance for large datasets.
- Tuned queries and integrated data using Python, REST APIs, and Zscaler, delivering insights for hundreds of daily users.

Relevant Projects

Real-Time Trading System—Event-Driven Clearance and Settlement Pipeline [github](#)

- Built real-time financial transaction processing system using Kafka, PostgreSQL, and Redis for sub-second clearance and settlement,
- Developed Python microservices for transaction validation, risk checks, and automated settlement processing
- Orchestrated containerized distributed systems using Docker Compose to enable robust, maintainable data pipeline deployments.
- Achieved real-time settlement status lookups and automated notifications to trading systems with Redis caching

GutMeal—Unlock the power of Gut [github](#)

- Developed a scalable Spark ML pipeline on Databricks for personalized meal plan prediction leveraging gut microbiome, lifestyle, and allergy data, achieving up to 94% accuracy.
- Utilized Spark SQL and PySpark MLlib to implement and benchmark multiple machine learning models including Decision Tree, Logistic Regression, Random Forest, and XGBoost on large-scale health datasets.
- Designed and maintained big data architecture with PySpark, minimizing Pandas usage to ensure efficient processing and resource use
- Performed Spark-native feature engineering and class upsampling, enhancing model precision while maintaining big data scalability.

Uber Data Pipeline—Star Schema and ETL Automation

- Designed a star schema data model and scalable ETL architecture for Uber trip data, enabling efficient analytics on big query.
- Automated data pipelines using Mage and Airflow on GCP, ensuring reliable ingestion, scheduling, and data freshness.
- Performed advanced SQL analytics to uncover trends, passenger behaviors, and trip patterns, driving data-driven insights.
- Integrated GCP Looker dashboards to visualize key metrics, empowering stakeholders with interactive, actionable reporting.